

homing guidance systems are sometimes called bi-static and mono-static systems, respectively.

RBS-15 anti-ship missile from Saab Bofors Dynamics, MBDA Exocet anti-ship and MBDA MICA surface-to-air and air-to-air missiles from MBDA, AS-34 Kormoran anti-ship missile from EADS and Indian DRDO Astra BVRAAM air-to-air missile (Fig. 12) are some examples of missiles that use active radar homing in the terminal phase.

Passive homing guidance makes use of some form of energy emitted by the target. This energy is intercepted by the missile seeker, which is processed to extract guidance information to guide the missile to home on to the target. This energy could be in the form of heat energy generated by the target, which is made use of by the seeker in an infra-red-guided missile. Infra-red-guided missiles constitute an important category of electro-optically-guided precision-strike weapons.

Anti-radiation missiles, such as AGM-88 HARM air-to-ground missiles, track the RF energy emitted by ground based radar stations to generate guidance signals. Passive torpedoes make use of sound waves generated by the engines of ships or sonars to attack their targets. There are missiles such as AGM-65 Maverick that are equipped with electro-optic sensors that rely on visual images to guide the weapon to the target.

In the case of re-transmission homing guidance, the target is illuminated by external radar. The energy reflected by the target is intercepted by the missile sensor. In this case, the missile does not have an onboard computer to process the sensor signal and generate guidance command.

Instead, the sensor signal is transmitted back to the launch platform for processing. Command signals generated at the launch platform are re-transmitted back to the missile for use by the missile's control surfaces to guide the missile to home on to the target.



Fig. 13: MIM-104 Patriot surface-to-air missile system



Fig. 14: AMRAAM missile

The advantage with this guidance technique, also called track-via-missile, is that the expensive tracking and processing hardware is reusable and does not get destroyed along with the missile. But, it requires a high-speed communication link between the missile and launch station. MIM-104 Patriot surface-to-air missile system (Fig. 13) of Raytheon Company of the USA is an example.

Navigation guidance. The term guidance not only refers to the determination of the desired path of travel (also called trajectory) from the vehicle's current location to an intended target, it also refers to the desired changes in velocity, rotation and acceleration needed to be executed for following the desired path.

The term navigation refers to the determination, at a given time, of the vehicle's present-state vector defined by location, velocity and attitude. The term control refers to the manipulation of the forces, by way of steering controls, thrusters, etc,

needed to track guidance commands while maintaining vehicle stability. These three functions are collectively known as navigation guidance.

Navigation guidance is further sub-divided into inertial navigation, ranging navigation, celestial navigation and geophysical navigation.

In the case of inertial navigation guidance, the vehicle uses onboard sensors to determine its motion and acceleration with the help of inertial measurement unit (IMU) or inertial navigation sensor (INS). This system works by telling the vehicle where it is at the time of launch, and the vehicle's computer uses the signals from the inertial measurement unit to ensure that the vehicle travels along the programmed path. Inertial navigation systems are widely used on a range of aerospace vehicles, which include commercial airliners, military aircraft and spacecraft.

With reference to precision-guided munitions, navigation guidance is used for mid-course correction of guided missiles. Long-range all-weather subsonic cruise missile Tomahawk and medium-range all-weather beyond-visual-range air-to-air missile AMRAAM (Fig. 14) are some examples that use inertial navigation for mid-course guidance.

While inertial navigation guidance technique makes use of onboard sensors, ranging navigation depends on external signals for guidance, which are usually provided by radio beacons. Based on the direction and strength of the signals received by the aircraft, it navigates its way along the desired trajectory.

Ranging navigation guidance has been largely rendered obsolete with the arrival of global positioning system (GPS). GPS based navigation has largely replaced radio beacons in both military and civilian applications. GPS is a key enabling technology for existing and future military precision navigation applications.

Joint direct attack munitions (JDAM) series of guided bombs make use of integrated INS and GPS guid-